

API Data Collection

Lesson 08 · ISBA 4715

Agenda

	Part	What
1	API Concepts	What APIs are and why they matter (<i>slides</i>)
2	Weather API by Hand	Your first API call (<i>live code</i>)
3	Loops + Bulk Collection	Getting enough data (<i>AI-assisted</i>)
4	Your Project Connection	Finding APIs for your domain

What apps do you use?

- Instagram
- Uber / Venmo
- ChatGPT / Spotify



APIs Are Everywhere

When you check the weather on your phone → **API call**

When Uber charges your card → **API call**

When Instagram loads your feed → **API call**

When ChatGPT answers a question → **API call**

The apps you use are wrappers around other services, connected by APIs.



API = Restaurant

Role	What They Do
You	Place an order
Waiter	Carry it to the kitchen
Kitchen	Prepare your food
Waiter	Bring back your meal

You never enter the kitchen.

You don't need to know how it works.

What Is an API?

| A remote service that lets you request data or trigger actions in another system

You send a request:

"Give me the current weather for Los Angeles"

You get back a response:

```
{ "temp_f": 72.0, "condition": "Sunny" }
```

REST: Request + Response

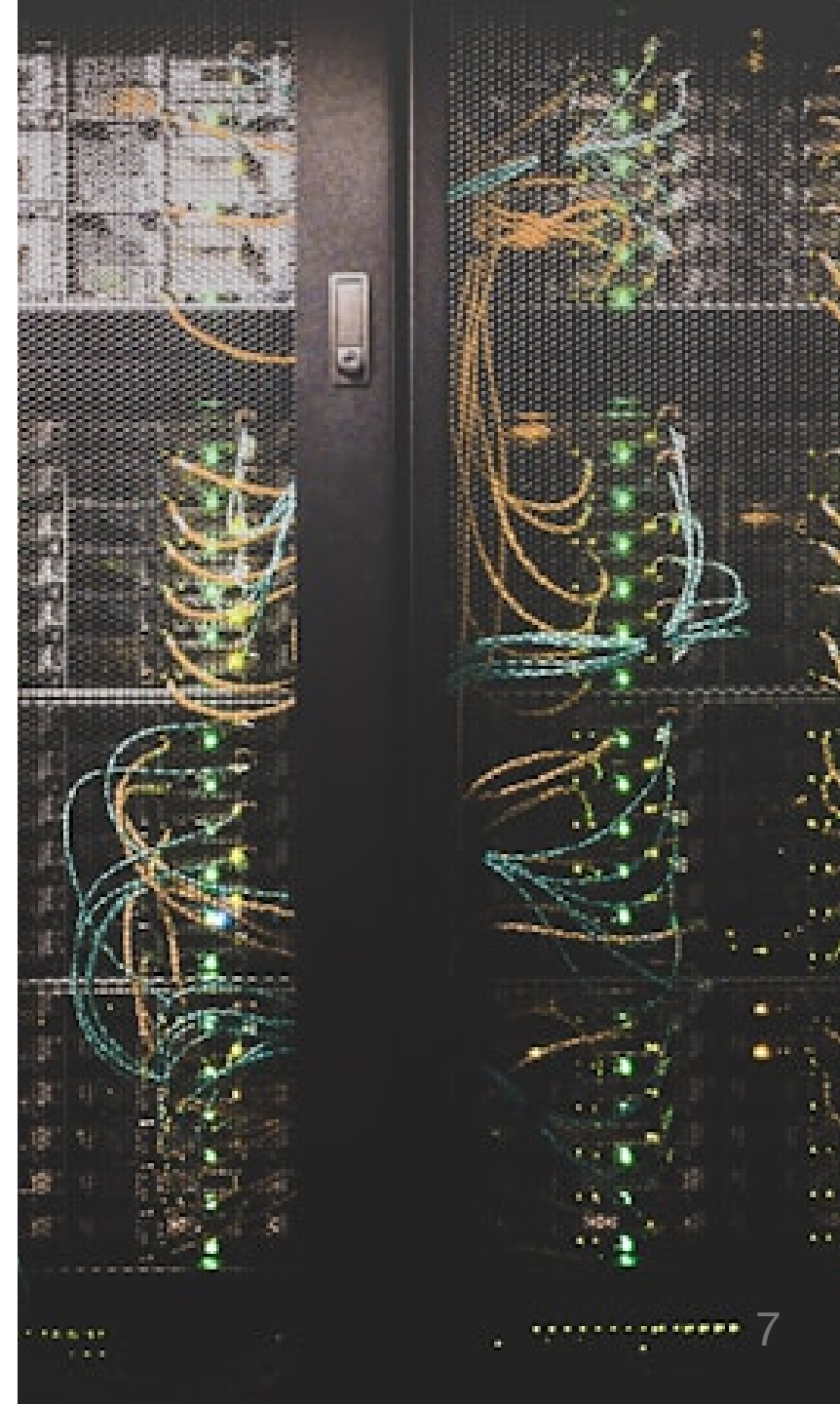
Two actions:

- **GET** = "Give me data"
- **POST** = "Create something"

Three parts to a request:

- **URL** — where to send it
- **Parameters** — what you want
- **API key** — who you are

Response comes back as **JSON**.



JSON: The Language of APIs

APIs return data in JSON — a structured text format:

```
{  
  "location": {  
    "name": "Los Angeles",  
    "region": "California"  
  },  
  "current": {  
    "temp_f": 72.0,  
    "condition": {  
      "text": "Sunny"  
    }  
  }  
}
```

{ } = object [] = list Data is **nested** — objects inside objects

API Keys

- APIs require a **key** to identify you
- Free tier = limited calls per day
- **Never commit keys to GitHub**

Store keys in `.env` files.

Think of the key as a library card:
it tracks usage and enforces limits.



Build vs. Buy

Need	Don't Do This	Do This
Weather data	Build a weather station	Call the Weather API
Music data	Scrape Spotify's website	Call the Spotify API
Financial data	Download CSVs manually	Call a market data API

APIs let you access data without building the source yourself.

One API Call = One Data Point

```
GET /current.json?q=90045
```

gives you weather for **one** zip code, **right now**

That is **one row** in your dataset.

Your project needs hundreds of rows. How?

Loops = Datasets

20 locations → Weather for 20 cities → **20 rows**

× **7 days each** → Weekly forecast per city → **140 rows**

× **90 days history** → Three months of data → **1,800 rows**

One API call = one row. **A loop = a dataset.**

The Pattern

request → parse → loop → DataFrame →
CSV

This works for any API: weather, financial, sports, music, government data.

Once you know it, switching to a new API is mostly just reading documentation.

Your Project Needs an API

- Milestone 01 requires an API source
- Today's pattern works for any REST API
- Your job posting tells you what data matters

Next up: code it together

